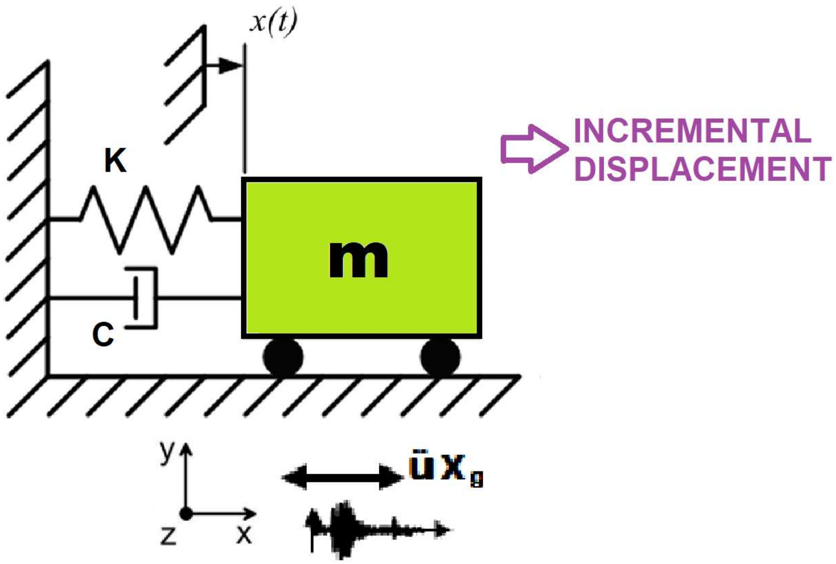
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES WITH AND WITHOUT PARALLEL COMPUTING

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

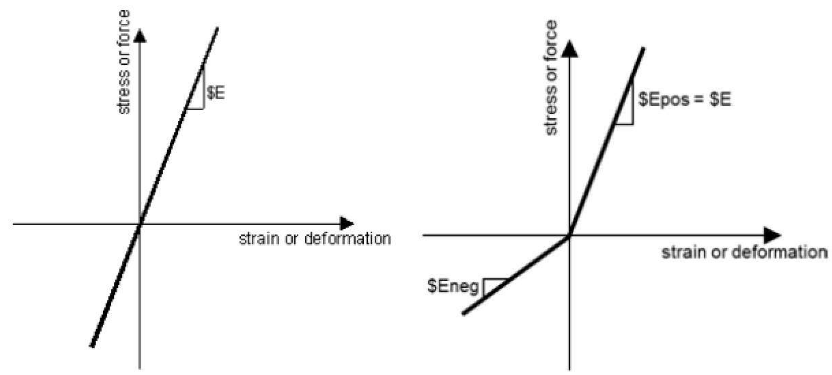
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

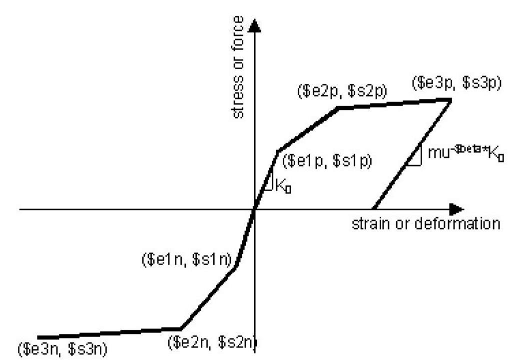
$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

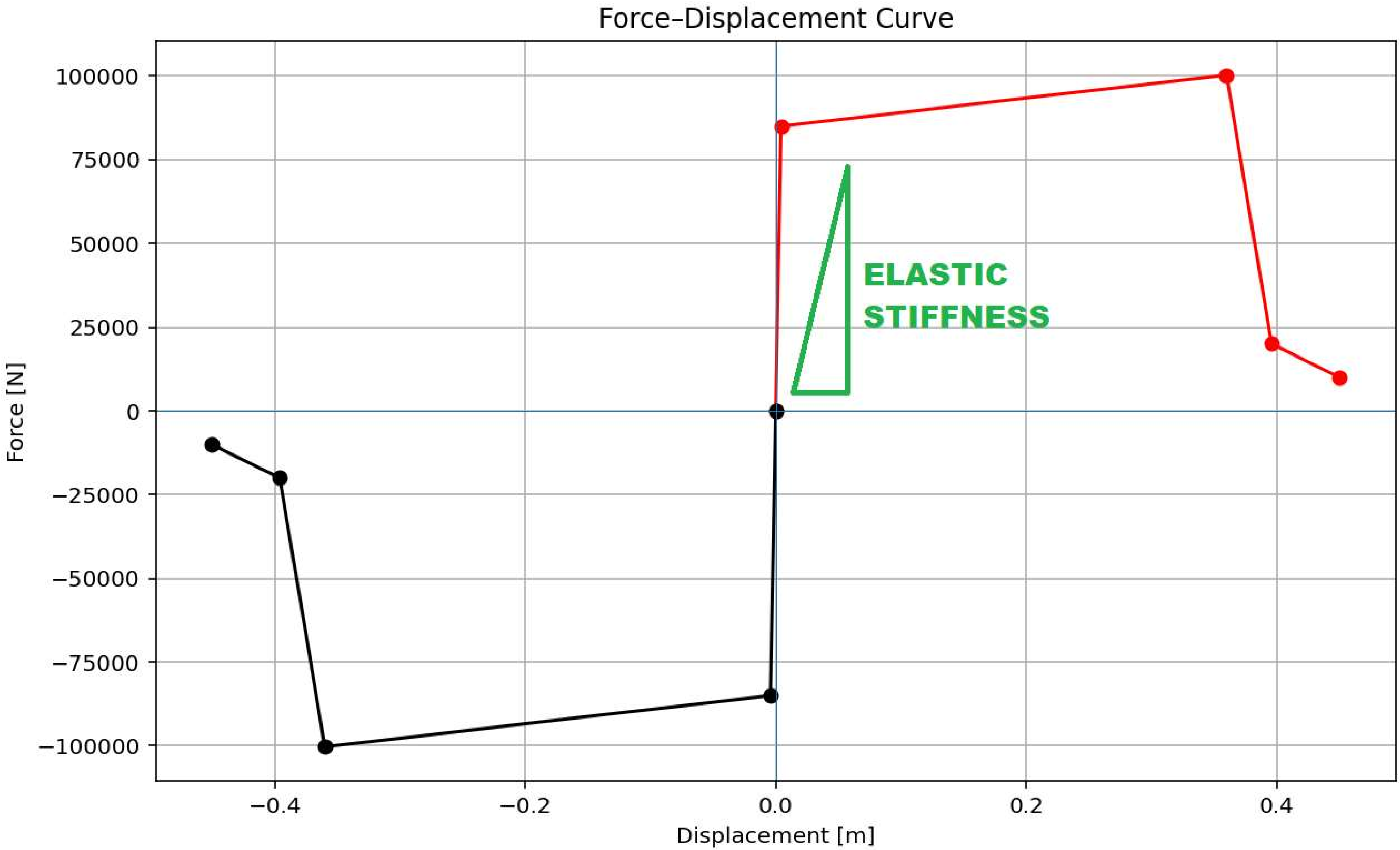
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL

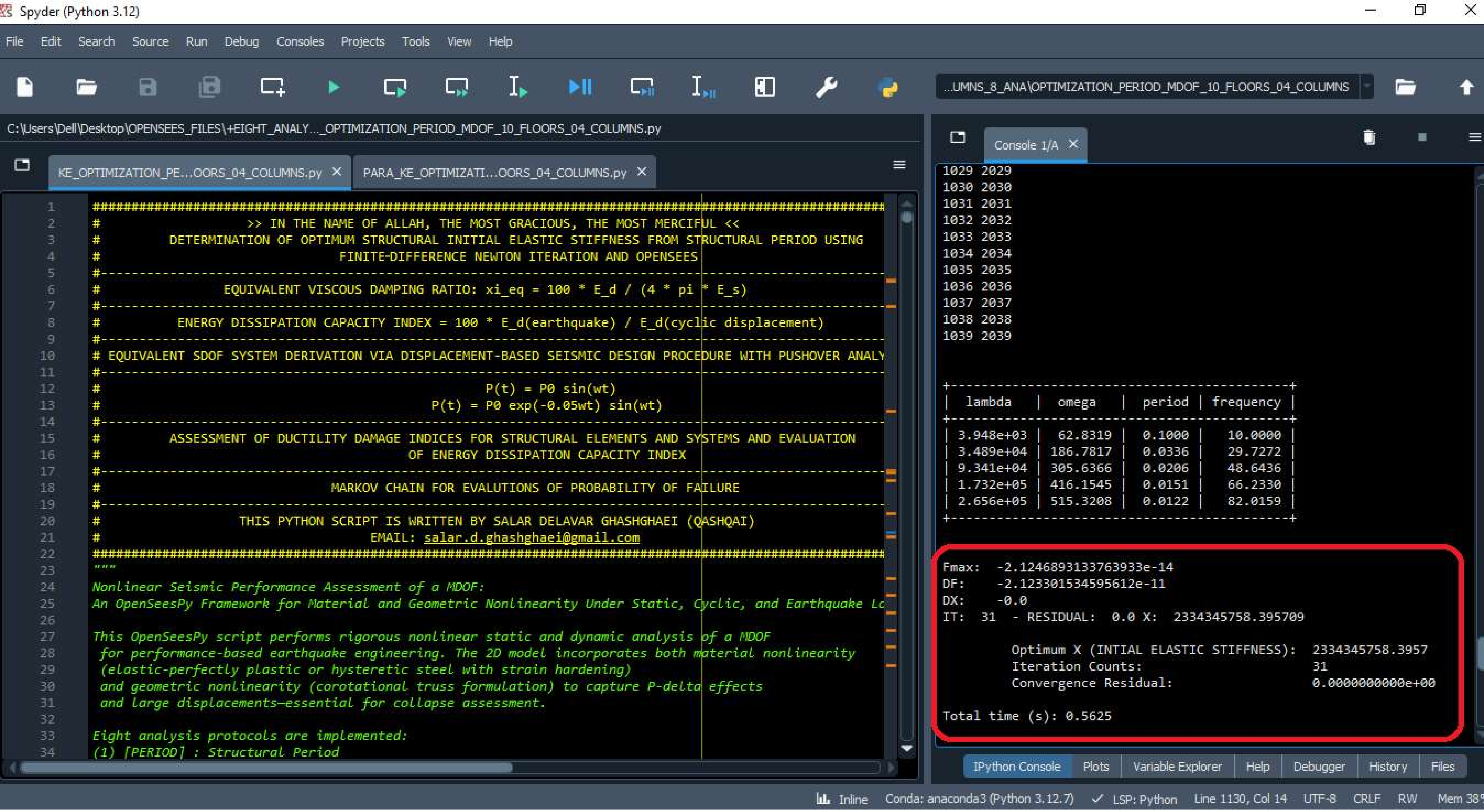


INELASTIC MATERIAL



Force–Displacement Curve
ELASTIC
STIFFNESS
Force [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4

OPTIMIZATION PROBLEM FOR STRUCTURAL PERIOD BY CHANGING INTIAL ELASTIC STIFFNESS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...UMNS_8_ANA\OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS.py
KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py
PARA_KE_OPTIMIZATI...OORS_04_COLUMNS.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING
FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALY
P(t) = P0 sin(wt)
P(t) = P0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
OF ENERGY DISSIPATION CAPACITY INDEX
MARKOV CHAIN FOR EVALUTIONS OF PROBABILITY OF FAILURE
THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a MDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Lo
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
 and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
Console 1/A
1029 2029
1030 2030
1031 2031
1032 2032
1033 2033
1034 2034
1035 2035
1036 2036
1037 2037
1038 2038
1039 2039
lambda	omega	period	frequency
3.948e+03	62.8319	0.1000	10.0000
3.489e+04	186.7817	0.0336	29.7272
9.341e+04	305.6366	0.0206	48.6436
1.732e+05	416.1545	0.0151	66.2330
2.656e+05	515.3208	0.0122	82.0159
Fmax: -2.1246893133763933e-14
DF: -2.123301534595612e-11
DX: -0.0
IT: 31 - RESIDUAL: 0.0 X: 2334345758.395709
Optimum X (INTIAL ELASTIC STIFFNESS): 2334345758.3957
Iteration Counts: 31
Convergence Residual: 0.0000000000e+00
Total time (s): 0.5625
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1130, Col 14 UTF-8 CRLF RW Mem 38%

Fmax: -2.1246893133763933e-14
DF: -2.123301534595612e-11
DX: -0.0
IT: 31 - RESIDUAL: 0.0 X: 2334345758.395709

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Convergence Residual: 0.0000000000e+00

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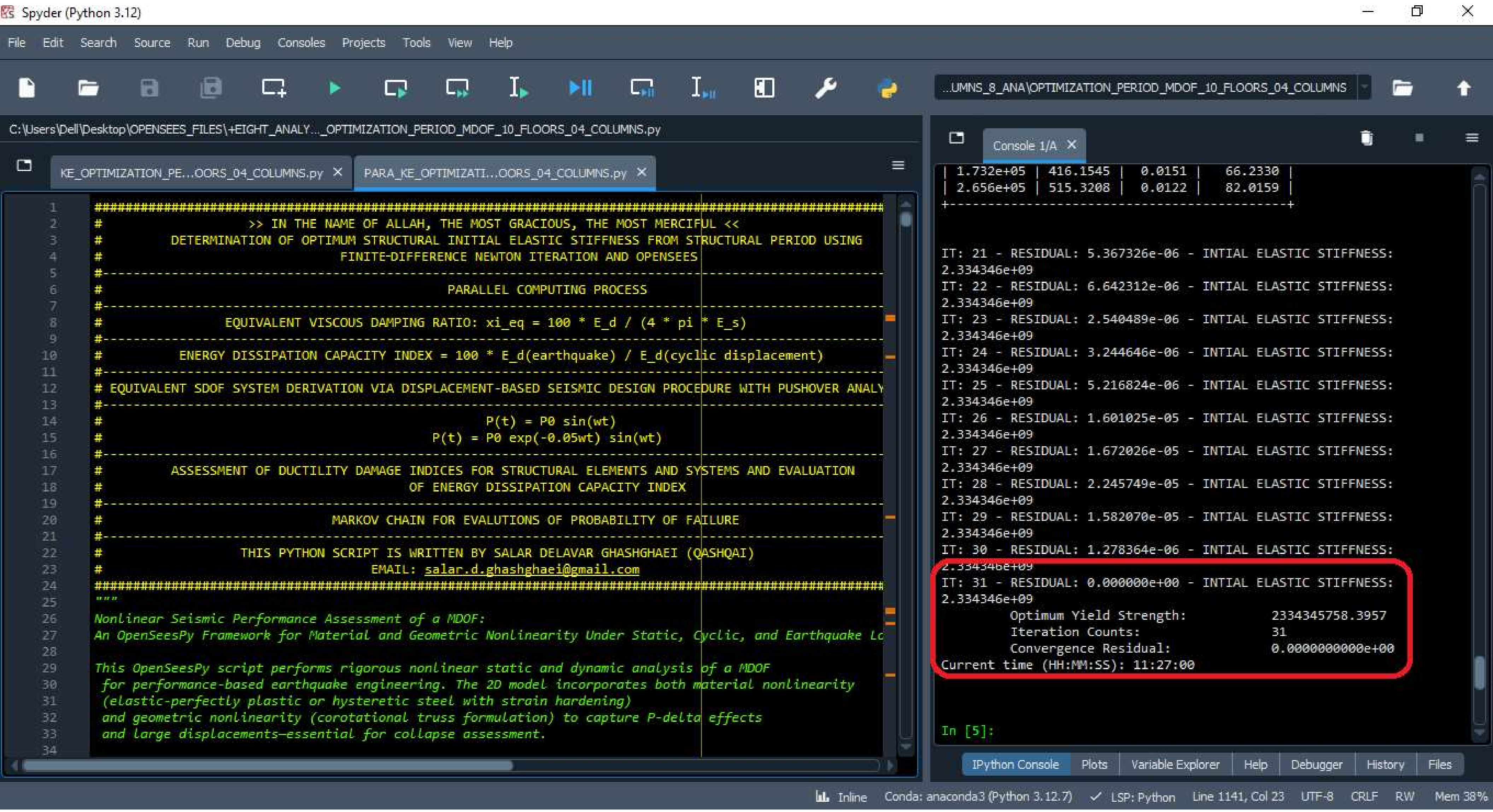
OPTIMIZATION PROBLEM FOR STRUCTURAL PERIOD BY CHANGING INTIAL ELASTIC STIFFNESS

(PARALLELCOMPUTING)


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...UMNS_8_ANA\OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS.py
KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py
PARA_KE_OPTIMIZATI...OORS_04_COLUMNS.py
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and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
and large displacements—essential for collapse assessment.
Console 1/A
| 1.732e+05 | 416.1545 | 0.0151 | 66.2330 |
| 2.656e+05 | 515.3208 | 0.0122 | 82.0159 |
IT: 21 - RESIDUAL: 5.367326e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 22 - RESIDUAL: 6.642312e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 23 - RESIDUAL: 2.540489e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 24 - RESIDUAL: 3.244646e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 25 - RESIDUAL: 5.216824e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 26 - RESIDUAL: 1.601025e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 27 - RESIDUAL: 1.672026e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 28 - RESIDUAL: 2.245749e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 29 - RESIDUAL: 1.582070e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 30 - RESIDUAL: 1.278364e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 31 - RESIDUAL: 0.000000e+00 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
Optimum Yield Strength: 2334345758.3957
Iteration Counts: 31
Convergence Residual: 0.0000000000e+00
Current time (HH:MM:SS): 11:27:00
In [5]:
IPython Console Plots Variable Explorer Help Debugger History Files
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1141, Col 23 UTF-8 CRLF RW Mem 38%

IT: 31 - RESIDUAL: 0.000000e+00 - INTIAL ELASTIC STIFFNESS: 2.334346e+09

Optimum Yield Strength: 2334345758.3957

Iteration Counts: 31

Convergence Residual: 0.0000000000e+00

Current time (HH:MM:SS): 12:03:41